

PRIYAM PANDEY

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RESEARCH SUMMARY

AI researcher working at the intersection of multimodal representation learning, 3D geometric deep learning, and biological foundation models. My research focuses on learning robust representations from heterogeneous biological modalities, developing adaptive and confidence-aware fusion mechanisms, and systematically evaluating how learned representations capture structural, functional, and biological information.

RESEARCH EXPERIENCE

- **DIL Lab, IISER Thiruvananthapuram** 2025 – Present
Undergraduate Researcher, advised by Alwin Poulose Thiruvananthapuram, Kerala, India
 - Conducted research in multimodal representation learning and biological AI under the supervision of **Alwin Poulose**.
 - Research contributed to a **first-author publication in Knowledge-Based Systems** and a **first-author manuscript under review in Neurocomputing**.

PUBLICATIONS & PREPRINTS

- **Multimodal Learning With Thermal Imagery and Pose Cues for Privacy-Preserving Behavior Recognition** 2026
Priyam Pandey, Alwin Poulose
Knowledge-Based Systems (Elsevier) | **First Author** | DOI: [10.1016/j.knosys.2026.116659](https://doi.org/10.1016/j.knosys.2026.116659)
 - Developed Confidence-Gated Cross-Attention (CGCA) and Cross-Gated Hierarchical Cross-Attention (CGHCA) for confidence-aware multimodal representation fusion.
 - Investigated adaptive modulation of cross-modal information flow based on local observational confidence across modalities.
 - Systematically evaluated the framework using 5-seed statistical validation to assess robustness and reliability.
- **Morphogenesis Fusion: Multi-View Representation Learning for Neuronal Morphology Analysis** 2026
Priyam Pandey, Alwin Poulose
Under Review in Neurocomputing (Elsevier) | **First Author**
 - Developed a tri-modal geometric representation-learning framework integrating 3D point clouds, graph representations, and volumetric CNN features.
 - Introduced Morphogenesis Fusion to combine complementary geometric and topological representations for 3D biological morphology.
- **Adaptive Multimodal Protein Foundation Models for Unified Protein Representation Learning (ProteinFM)** 2026
Priyam Pandey
Manuscript in Preparation / Active Research Project
 - Developing PAEP (Parameter-Efficient Adapter-Encoder-Projector) for aligning heterogeneous protein foundation-model representations.
 - Developing APF (Adaptive Poly-modal Fusion) for integrating sequence/evolutionary and structural representations through adaptive, confidence-aware fusion.
 - Developing a 9-stage U_0 – U_8 probing framework for systematic characterization of protein representations.

MAJOR RESEARCH PROJECTS

- **ProteinFM: Adaptive Multimodal Protein Foundation Models** 2026 – Present
Core Area: Protein Foundation Models & Biological Representation Learning Active Research Project
 - **Representation Alignment:** Developing model-agnostic PAEP adapters to map heterogeneous protein foundation-model representations into a shared latent representation space.
 - **Multimodal Fusion:** Developing APF for adaptive integration of sequence/evolutionary and structural representations, incorporating reliability signals such as pLDDT.
 - **Protein Foundation Models:** Working with protein representation models including ESM-2, ProtT5, SaProt, and GearNet.
 - **Representation Probing:** Developing the U_0 – U_8 evaluation framework to characterize representation quality across biological similarity, functional properties, mutation-fitness prediction, structural geometry, transferability, representation geometry, and computational efficiency.

- **Evaluation Rigor:** Incorporating explicit data-leakage controls, unimodal and multimodal baselines, component-level ablations, and multi-seed evaluation into the benchmarking framework.
- **NMC RL: Multiview Geometric Deep Learning for 3D Biological Morphometry** 2025 – 2026
Core Area: Geometric Deep Learning & 3D Biological Representation Learning *Under Review, Neurocomputing*
 - **Multi-View Representation:** Developed tri-modal architectures representing biological morphology through 3D point clouds, graph structures, and volumetric representations.
 - **Geometric Learning:** Used PointNet++, graph neural networks including PNA/GAT, and 3D CNNs to capture complementary spatial and topological information.
 - **Morphogenesis Fusion:** Developed a hierarchical fusion mechanism for integrating representations from different geometric views of biological morphology.
- **Confidence-Gated Multimodal Representation Learning** 2025 – 2026
Core Area: Multimodal Attention & Uncertainty-Aware Learning *Published, Knowledge-Based Systems*
 - **Confidence-Aware Attention:** Developed CGCA/CGHCA to modulate cross-modal information flow according to modality-specific observational confidence.
 - **Representation Fusion:** Investigated the incorporation of modality reliability into multimodal attention mechanisms.
 - **Robustness Evaluation:** Evaluated multimodal fusion under noise and heterogeneous modality reliability using a disciplined multi-seed experimental protocol.

RESEARCH INTERESTS

- **Biological Foundation Models & Structural AI:** Protein foundation models, multimodal protein representation learning, protein sequence–structure representation learning, differentiable computational structural biology.
- **Geometric Deep Learning & Fusion:** 3D geometric deep learning, structural representation learning, geometric and structural modeling of biological systems, adaptive and confidence-aware multimodal fusion.
- **Representation Theory & Evaluation:** Biological representation learning, foundation-model probing and systematic representation evaluation frameworks (U_0 – U_8).

EDUCATION

- **Indian Institute of Science Education and Research Thiruvananthapuram** 2024 – Present
BS-MS Dual Degree, School of Data Science (Current Semester: 5th) Kerala, India
 - **Relevant Coursework:** Mathematical Statistics, Optimization Techniques, Data Structures, Machine Learning I & II, Applied Regression Analysis, Numerical Analysis, Scientific Computing, Database Management Systems, Computer Organization & Operating Systems.
 - **Academic Focus:** Machine learning, statistical modeling, mathematical foundations of data science, scientific computing, and biological representation learning.